

Reviving Online Sales: Addressing Order Decline in a Biryani Cloud Kitchen

A Project Proposal Report

Name: Sharmila L

Roll number: 21f20XXXXX



*IITM BS Degree Program,
Indian Institute of Technology, Madras,
Chennai Tamil Nadu, India, 600036*

Table of contents

Reviving Online Sales: Addressing Order Decline in a Biryani Cloud Kitchen.....	1
Table of contents.....	2
Declaration Statement.....	3
1. Executive Summary.....	4
2. Organization Background.....	4
3. Problem Statement.....	5
3.1 Lack of Insight into Year-on-Year Growth.....	5
3.2 Fluctuations in Online Orders Impacting Business Sustainability.....	5
3.3 Low Repeat Customer Rate Affecting Customer Base Growth.....	5
4. Background of the Problem.....	5
5. Problem Solving Approach.....	6
5.1 Details about methods.....	6
5.2 Data Collection Process.....	6
5.3 Tools.....	7
6. Expected Timeline.....	7
7. Expected Outcomes.....	8

Declaration Statement

I am working on a Project titled “**Reviving Online Sales: Addressing Order Decline in a Biryani Cloud Kitchen**”. I extend my appreciation to Mrs. Rigxxxx Paxxxx, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

XXXXXX

Signature of Candidate:

Name: Sharmila

Date: 08/02/2025

1. Executive Summary

Cloud Kitchen Business, located in Othakadai, Madurai, specializes in Chennai-style Basmati Biryani. It is a privately owned cloud kitchen founded by Mrs. Rixxxx Pxxxxx, a passionate woman entrepreneur, in April 2023. Her vision is to grow Ax-Wxxx Bxxxxxx into a well-recognized brand, similar to Behrouz Biryani. While the business primarily operates as a cloud kitchen, they also do bulk party orders.

Recently, the business is facing challenges with **stagnant growth** and has observed a decline in online orders, despite offering the lowest pricing and maintaining quality service. The lack of repeat customer details and limited visibility in the highly competitive Madurai food market have further impacted its sales performance.

To understand key issues and challenges, the project will analyze customer trends, order history, and competitor positioning including sales trend evaluation, menu performance study, and brand image. The expected outcome is to identify actionable strategies that will help Ax-Wxxx Bxxxxxx **improve online engagement, retain customers**, and position itself for sustained growth.

2. Organization Background

[Ax-Wxxx Bxxxxxx](#) is a cloud kitchen operating since April 2023 at Othakadai, Madurai. It was founded by Mrs. Righana Parveen, a passionate entrepreneur. The idea for the cloud kitchen business stemmed from her family's struggle to find **authentic Chennai-style Biryani** in Madurai, where other varieties like **Seeraga Samba Biryani and Dindigul Biryani dominate the market**. This gap inspired her to bring Chennai's signature flavour to the city.

With a focused menu, their hero item, Regular Chicken Biryani, is priced at just ₹80 to ensure accessibility while maintaining rich taste. Operating daily from 11:30 AM to 5 PM, Ax-Wxxx Bxxxxxx runs efficiently with a small team of four employees and a master chef with a five-seat capacity for dine-in. Their main business is through Swiggy and Zomato deliveries. Additionally, they cater to offline bulk party orders across India, allowing customers to experience their flavorful biryani beyond Madurai.

3. Problem Statement

3.1 Lack of Insight into Year-on-Year Growth: Ax-Wxxx Bxxxxxx has not observed significant growth as expected, and the factors remain unclear. This project's focus is to analyze sales to identify key areas of improvement and growth.

3.2 Fluctuations in Online Orders Impacting Business Sustainability: To improve online order volume, this project seeks to develop strategies to enhance online visibility, optimize platform presence, and improve customer engagement.

3.3 Low Repeat Customer Rate Affecting Customer Base Growth: Currently, there is no structured strategy followed for customer retention. This project aims to analyze order patterns, understand the factors influencing repeat purchases, and develop targeted solutions.

4. Background of the Problem

The biggest challenge seems to be the **unknown customer base** for Chennai-style Biryani in a market where Seeraga Samba and Dindigul style Biryani reign supreme. When the business first launched, it received an overwhelming response, with orders pouring in. However, over the past year, growth has slowed down, and online orders have not picked up as expected. Despite maintaining competitive pricing and offering discounts, the business has not seen a significant increase in online sales. While their offline bulk catering orders are doing well solely through word-of-mouth marketing, online customer acquisition remains a challenge.

Business's limited online visibility and gaps in customer awareness have made attracting and retaining customers a real struggle. Externally, Swiggy and Zomato's algorithm challenges like unexpected app-triggered promotions without owner consent, fluctuating delivery partner availability, and strong regional preferences for Seeraga Samba and Dindigul Biryani further affect growth. Additionally, unclear platform commission structures create pricing constraints, limiting flexibility in promotional strategies. This project aims to identify insights to address these challenges and develop proactive strategies to help increase online sales while aligning with the business's long-term growth aspirations.

5. Problem Solving Approach

5.1 Details about methods

To understand the reasons behind Ax-Wxxx Bxxxxxx's slow growth, this project will use **Company, Competitor, and Customer (3Cs) Framework**.

The first step is Company Analysis through SWOT assessment, gathering insights from discussions with owner and operational data. Next is Competitor Analysis, to understand market positioning, pricing strategies, and service gaps, allowing Ax-Wxxx to identify competitive threats and potential opportunities. Customer Analysis will dive into historical order data, studying purchasing patterns, conversion rates, and customer behaviour, with a particular focus on distinguishing new vs. repeat customers to study retention.

Further, analysis of **Sales Trend** through Exploratory Data Analysis (EDA) will examine monthly growth, average order value (AOV), and seasonal variations to detect revenue fluctuations. Lastly, **Menu Performance Analysis** will assess item popularity to refine offerings and pricing strategies for better customer engagement. Together, these structured assessments will provide data-backed insights to enhance Ax-Wxxx's online presence, improve customer acquisition, and align sales growth with long-term business objectives.

5.2 Data Collection Process

For data collection, order history from Swiggy/Zomato will be the primary source. Additionally, insights will be gathered through interviews with the owner, and competitor information will be obtained from search results on delivery apps. The approach is categorized as follows:

Quantitative Data: Online Order History from Swiggy and Zomato will help track order trends, peak hours, and the popularity of menu items. Customer Data will be examined to distinguish between repeat and new customers, allowing for a better understanding of purchasing patterns and the customer journey within the ordering process. Additionally, if offline bulk order records are maintained, Offline Sales Data will provide valuable comparisons between online and offline business performance. Lastly, Search Ranking Data will be collected from food delivery app search results to study competitor rankings, reviews, and ratings, offering insights into online visibility and potential areas of improvement.

Qualitative Data: Interviews with the business owner and team will help understand operational challenges, strategies, and decision-making processes. Menu Cards will be analyzed to evaluate offerings, pricing strategies, and customer preferences for specific dishes. Additionally, packaging details will be assessed to understand the branding, presentation, and overall appeal of food packaging, ensuring that it aligns with customer expectations and market standards.

5.3 Tools

The analysis will be carried out using Python and its libraries like Matplotlib for trend identification, segmentation, and visualization. These tools provide a wide range of options for detailed analysis and my familiarity in using this tool is another key reason for choosing these tools.

Through the above-mentioned approaches, the project aims to provide a clear understanding of business gaps and actionable recommendations to improve Ax-Wxxx's online sales performance and overall growth.

6. Expected Timeline

6.1 Work Breakdown Structure (WBS)

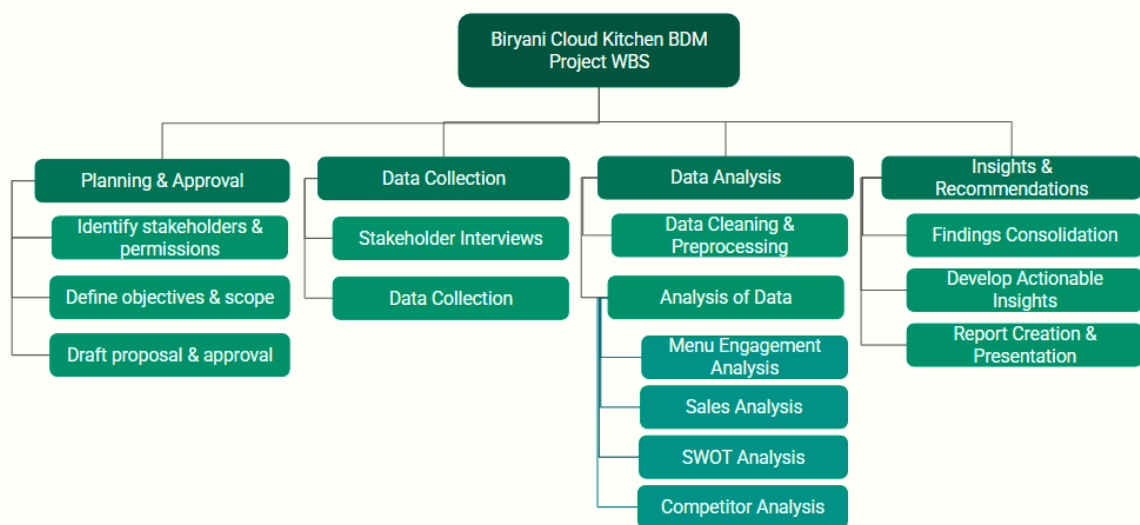


Figure 1 - Work Breakdown Structure

6.2 Gantt Chart

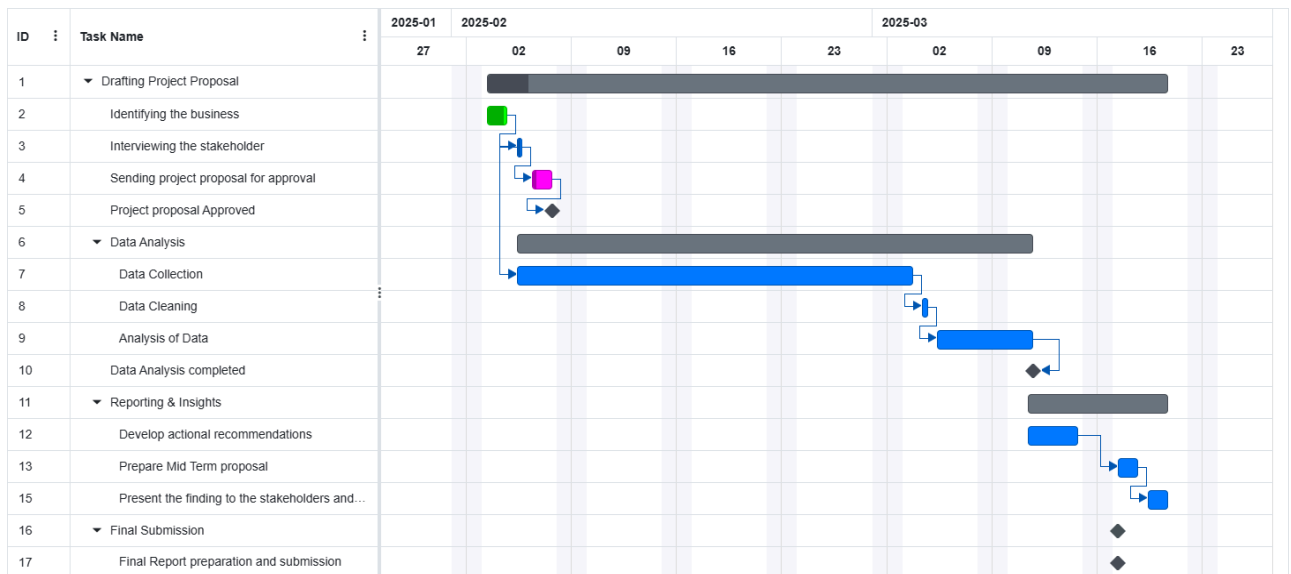


Figure 2 - Project Workflow - Gantt Chart

The project is estimated to take approximately 30 to 45 days, following the workflow outlined in Figure 1 and Figure 2.

7. Expected Outcomes

The goal is to provide **data-driven insights** that will help Ax-Wxxx Bxxxxxx **overcome its day to day challenges**. By analyzing relevant data, the project is expected to provide:

- The key factors behind decreasing online sales.
- A deeper understanding of customer behaviour to identify their purchasing patterns and preferences.
- Menu optimization strategies to enhance customer conversions and boost order value.
- Strategies to build long-term customer relationships and improve retention.
- Actionable branding strategies to strengthen market presence and drive business growth.